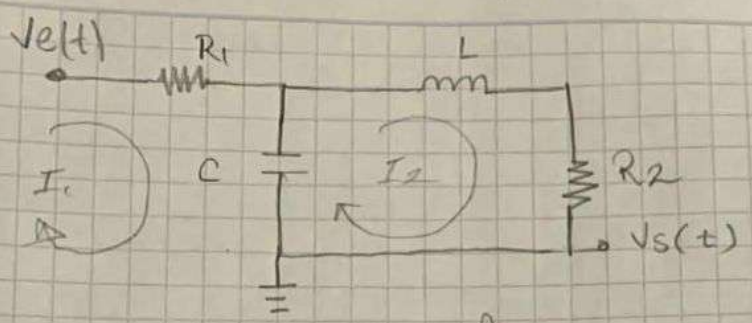

$$Ve(t) = R_1 \cdot I_1(t) + \frac{1}{C}\int (I_1 - I_2)\,dt$$

$$\frac{1}{C}\int (I_1 - I_2)\,dt = L\frac{dI_2(t)}{dt} + R_2 I_2(t)$$

$$Vs(t) = R_2 I_2(t)$$

Pasamos al dominio de laplace

- $Ve(s) = R_1 \cdot I_1(s) + \frac{1}{C}(I_1(s) - I_2(s))$
- $\frac{1}{Cs}(I_1(s) - I_2(s)) = Ls\,I_2(s) + R_2 I_2(s)$
- $Vs(s) = R_2 I_2(s)$

Separamos I_1 e I_2

$$Ve(s) = R_1 \cdot I_1(s) + \frac{1}{C} I_1(s) - \frac{1}{C} I_2(s) = \left(R_1 + \frac{1}{C}\right) I_1(s) - \frac{1}{C} I_2(s)$$

Eq 1 → $Ve(s) = \left(R_1 + \frac{1}{C}\right) I_1(s) - \left(\frac{1}{C} \cdot I_2(s)\right)$

Eq 2 → $0 = Ls\,I_2(s) + R_2 I_2(s) + \frac{1}{Cs} I_1(s) + \frac{1}{Cs} I_2(s)$

$$0 = \left(Ls + R_2 + \frac{1}{Cs}\right) I_2(s) + \frac{1}{Cs} I_1(s)$$

Eq 3 → $Vs(s) = R_2 I_2(s)$

Despejo de I_1 en equ 2

$$\frac{1}{Cs} I_1(s) = \left(Ls + R_2 + \frac{1}{Cs}\right) I_2(s)$$

$$I_1(s) = \left(Ls + R_2 + \frac{1}{Cs}\right) I_2 \cdot Cs$$

$$I_1(s) = (CLs^2 + CR_2 s + 1)\, I_2(s)$$

$$V_e(s) = \left(R_1 + \frac{1}{C}\right)\left(CLs^2 + CR_2 s + 1\right)\left(I_2(s)\right) - \left(\frac{1}{C} \cdot I_2(s)\right)$$

$$V_e(s) \quad R_1 C L s^2 + (R_1 C R_2 + L)s + (R_1 + R_2) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{R_1 C L s^2 + (R_1 C R_2 + L)s + (R_1 + R_2)}$$